

Benchmarking DQN vs PPO on Atari Pong and Breakout

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I. Project Description

We compare **DQN** (Mnih et al., 2015) and **PPO** (Schulman et al., 2017) on **Pong** (easy baseline) and **Breakout** (harder test) using **pixel-based** environments (PongNoFrameskip-v4 , BreakoutNoFrameskip-v4) the same setting as the original papers. Both algorithms use an identical **Nature CNN** (3 conv + 2 FC) and preprocessing pipeline. Each (algo, env) combination runs with **5 seeds = 20 total runs** at 1M timesteps each. Evaluation: **100 deterministic episodes**, reporting mean and standard deviation ($\mu \pm \sigma$). **Deliverable** IEEE-format report, learning curves, tables, statistical analysis. Results'd be interpreted as fixed-budget comparison

II. State Space

Raw observations are Atari RGB frames of shape (210, 160, 3), represented as uint8 pixel values in [0, 255]. The final processed state is a normalized 4-frame stack of shape (4, 84, 84)

#	Operation	Output
1	Grayscale conversion	(210, 160, 1)
2	Resize to 84×84	(84, 84, 1)
3	Stack 4 frames	(4, 84, 84)
4	Normalize pixels and clip rewards	scaled pixels; rewards clipped to +/-1

III. Action Space

Environment	Action Space	Actions
Pong	Discrete(6)	NOOP, FIRE, RIGHT, LEFT, RIGHT+FIRE, LEFT+FIRE
Breakout	Discrete(4)	NOOP, FIRE, LEFT, RIGHT

Action repeat = 4. With Gymnasium ALE v5, use ALE/Pong-v5 and ALE/Breakout-v5 with explicit configuration, use PongNoFrameskip-v4 and BreakoutNoFrameskip-v4 with wrapper-controlled frame skip.

IV. Proposed Solution

Component	Choice
Framework	Stable Baselines3, PyTorch via TensorBoard, CSV log, 50k-step checkpoints
Environments	Gymnasium PongNoFrameskip-v4 , BreakoutNoFrameskip-v4
Algorithms	DQN, PPO - Canonical values from each paper and SB3 documentation
Policy	Nature CNN (3 conv - 2 FC, same for both)
Preprocessing	Grayscale - 84×84 → 4-frame stack - reward clip ± 1 via AtariWrapper
Timesteps	1,000,000 per run across 5 random seeds per (algo, env) = 20 runs
Evaluation	100 deterministic episodes, learning curves, reward/hour, ($\mu \pm \sigma$), effect size
Analysis	Learning curves, bar charts, Tukey HSD, reward/hour, seed-variance